

an m-stock portfolio executed over n trading horizons has only m parameters to determine regardless of the trading horizon whereas the complete problem and QP optimization each have decision variables. Second, since Equation 9.10 holds for all stocks we are ensured of completion of the order. Third, since $e^{-j\omega_i} > 0$ for all j we guarantee we adhere to our shrinking portfolio constraint. Most essential, however, is that since the residual schedule is expressed in terms of a continuous exponential function, the analytical gradient and Hessian can be easily computed. This dramatically increases the computational efficiency.

A limitation of the residual trade schedule exponential, similar to the exponential trade schedule, is that it does not allow as much freedom to take advantage of natural hedging and diversification as the exact NLP and QP approaches above. The lower and upper bounds are included on the trading schedule parameter to ensure the order is traded within a user specified rate.

Trading Rate Parameter

The trade strategy can also be expressed in terms of a trading rate parameter α . Here the number of shares to transact is equal to a specified percentage of market volume excluding the order shares. The process is best explained as follows. For a specified trading rate α , the expected time to complete the order (expressed as a percentage of a trading day) is:

$$t = \frac{X}{ADV} \cdot \frac{1}{\alpha}$$

If the trading day is segmented into n trading periods then the order will be completed in T periods where:

$$T = t \cdot n = \frac{X}{ADV} \cdot \frac{1}{\alpha} \cdot n \text{ (rounded up to the nearest integer)} \quad (9.12)$$

For example, if the order size $\frac{X}{ADV} = 10\%$ and the trading rate $\alpha = 10\%$ the order will complete in a day. If the trading rate is $\alpha = 20\%$ the order will complete in one-half day, and if the trading rate is $\alpha = 5\%$ the order will complete in two days.

Market Impact Expression

For a constant trading rate the temporary market impact cost for a single stock is:

$$MI(\alpha) = b_1 \cdot T^* \alpha$$